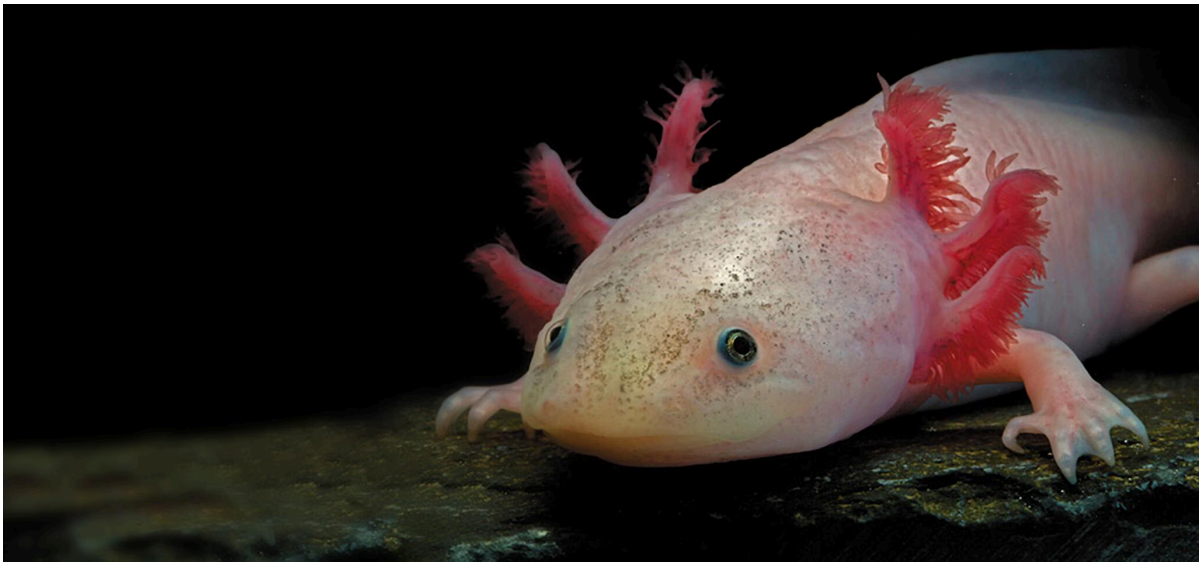
Chapter 42

Campbell Biology, AP Edition, 12e



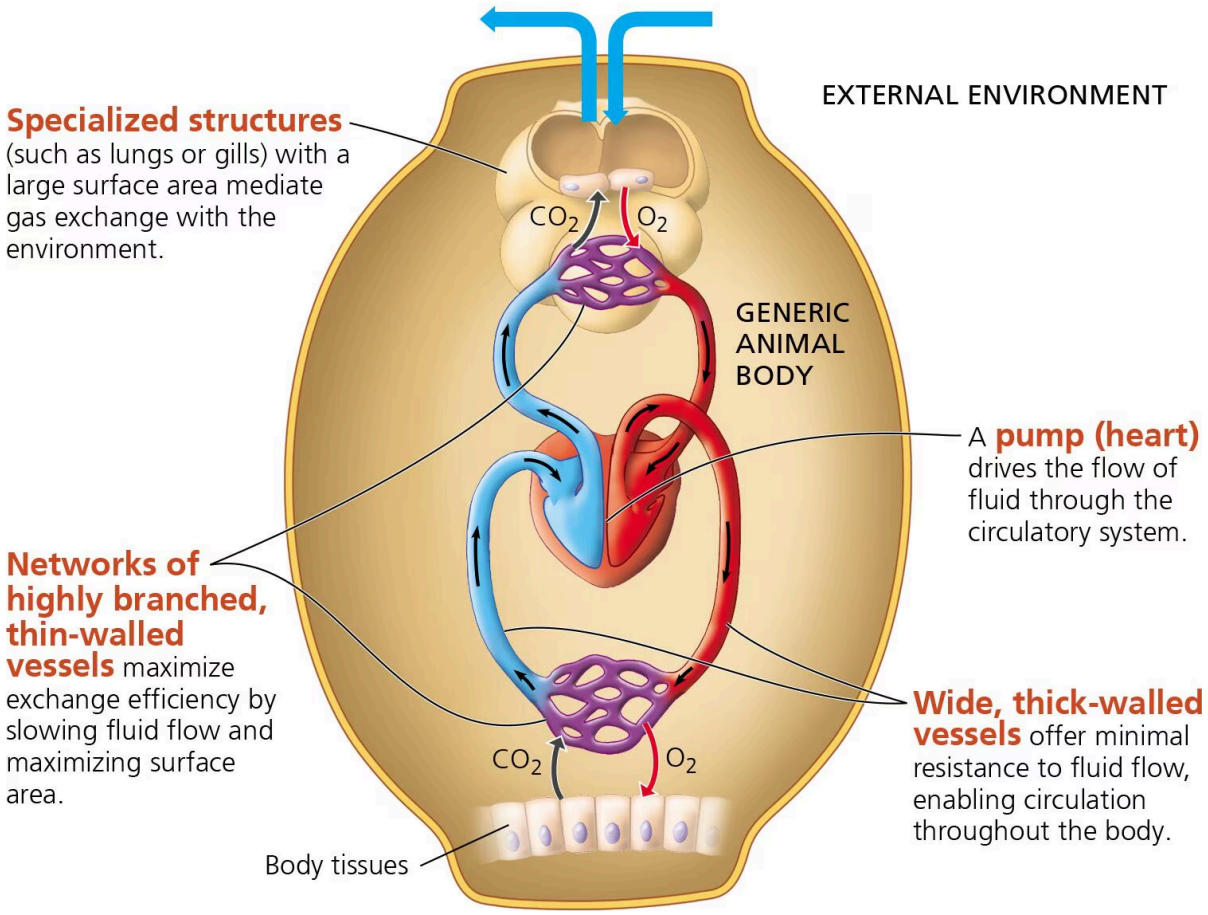
Chapter 42

Circulation and Gas Exchange

Figure 42.1

This animal may look like a creature from a science fiction film, but it's actually an axolotl, a salamander native to shallow ponds in central Mexico. The feathery red appendages jutting out from its head are gills. External gills, although uncommon in other adult animals, help axolotls carry out a process common to all organisms: the exchange of substances between body cells and the environment.

How are structure and function related in the exchange and circulation of oxygen and carbon dioxide?



A generic animal body is given. It has specialized structures (such as lungs or gills) with large surface areas that mediate gas exchange with the environment. A pump (heart) drives the flow of fluid through the circulatory system. Networks of highly branched, thin-walled vessels maximize exchange efficiency by slowing fluid flow and maximizing surface area. Wide, thick-walled vessels offer minimal resistance to fluid flow, enabling circulation throughout the body. O 2 enters, and C O 2 exits.

Key Concepts

- 42.1**Circulatory systems link exchange surfaces with cells throughout the body**
- 42.2**Coordinated cycles of heart contraction drive double circulation in mammals**
- 42.3**Patterns of blood pressure and flow reflect the structure and arrangement of blood vessels**
- 42.4**Blood components function in exchange, transport, and defense**
- 42.5**Gas exchange occurs across specialized respiratory surfaces**

- 42.6**Breathing ventilates the lungs**
- 42.7**Adaptations for gas exchange include pigments that bind and transport gases**

Organisms have various mechanisms for obtaining nutrients and eliminating wastes **(BIG IDEA 2)** that are more efficient with cooperative interactions **(BIG IDEA 4)**.

Study Tip

Draw and annotate a diagram: Sketch a simplified double circulation system like the idealized one on this page. Label where each of the following is highest: blood pressure, CO_2 concentration, O_2 concentration, and total cross-sectional area of the blood vessels. Note that the left side of your diagram represents the right side of the heart in the body, and remember "**R**e**V**i**T**a**L**ize" to help remind you of the path of blood from the **R**ight **V**entricle **t**o the **L**ungs.

Get Ready for This Chapter

In this chapter, you will learn how animals use circulatory systems to move nutrients and wastes between points in the body. You will also learn how animals use specialized gas exchange surfaces to transfer gases into and out of the body. Before beginning this chapter, you should be able to do the following:

- Identify the conditions under which diffusion can bring about the net movement of molecules across membranes (see Concept 7.3, Figure 7.11).
- Describe the relationships of O_2 and CO_2 to cellular respiration (see Concept 9.1 and Figure 9.1).
- Describe cooperative binding to a protein that has multiple subunits (see Concept 8.5, Figure 8.20).

The following questions provide a quick check of your basic knowledge in these areas.

1. Chambers A and B are separated by a membrane permeable to substance X. The concentration of X is 5 percent in A and 15 percent in B. The volume of A is three times that of B. Which of the following will occur?

Although diffusion is random, the result will be the net movement of X from B to A.

That's correct.

Correct! If you had trouble with this question, review the following material: Due to their thermal energy and thus constant motion, molecules undergo diffusion, spreading out into the available space. In the absence of other forces, the random movement of each individual molecule will cause a substance to diffuse from where it is more concentrated to where it is less concentrated, in this case from chamber B to chamber A.

There will be no net movement of X because the total amount of X is the same in A and B.

That's incorrect. Try again.

Incorrect. Review the following material: Due to their thermal energy and thus constant motion, molecules undergo diffusion, spreading out into the available space. In the absence of other forces, the random movement of each individual molecule will cause a substance to diffuse from where it is more concentrated to where it is less concentrated.

There will be no net movement of X because there is no energy available to transport X from B to A.

That's incorrect. Try again.

Incorrect. Review the following material: Due to their thermal energy and thus constant motion, molecules undergo diffusion, spreading out into the available space. In the absence of other forces, the random movement of each individual molecule will cause a substance to diffuse from where it is more concentrated to where it is less concentrated.

Because diffusion is random, diffusion will take place without changing the concentration of X in either A or B.

That's incorrect. Try again.

Incorrect. Review the following material: Due to their thermal energy and thus constant motion, molecules undergo diffusion, spreading out into the available space. In the absence of other forces, the random movement of each individual molecule will cause a substance to diffuse from where it is more concentrated to where it is less concentrated.

2. Cellular respiration

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uses oxygen and carbon dioxide as raw materials.
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That's incorrect. Try again.

Incorrect. Review the following material: Aerobic cellular respiration, which occurs in mitochondria, consumes oxygen and captures chemical energy that can be used to do work.

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produces carbon dioxide and ATP.
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That's correct.

Correct! If you had trouble with this question, review the following material: Aerobic cellular respiration occurs in mitochondria and is similar to the combustion of gasoline in an automobile engine, with energy captured to do work and carbon dioxide released as a waste product. In aerobic respiration, oxygen is consumed as a reactant along with the organic fuel, and chemical energy is captured in the form of ATP.

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takes place in specialized organelles called chloroplasts.
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That's incorrect. Try again.

Incorrect. Review the following material: Aerobic cellular respiration, which occurs in mitochondria, consumes oxygen and captures chemical energy that can be used to do work.

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synthesizes sugars from carbon dioxide and water.
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That's incorrect. Try again.

Incorrect. Review the following material: Aerobic cellular respiration, which occurs in mitochondria, consumes oxygen and captures chemical energy that can be used to do work.

3. As a property of enzymes, cooperativity refers to

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one enzyme subunit altering the behavior of others.
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That's correct.

Correct! If you had trouble with this question, review the following material: Cooperativity is a property of some enzymes with multiple subunits. Binding of a substrate molecule at one active site triggers a shape change at the remaining active sites that increases catalytic activity and thus amplifies the response of enzymes to substrates.

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two different enzymes using the same substrate.
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That's incorrect. Try again.

Incorrect. Review the following material: Cooperativity is a property of some enzymes with multiple subunits. It is triggered by the binding of a substrate molecule at one active site on the enzyme.

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one enzyme acting on two different substrates.
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That's incorrect. Try again.

Incorrect. Review the following material: Cooperativity is a property of some enzymes with multiple subunits. It is triggered by the binding of a substrate molecule at one active site on the enzyme.

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catalysis that consumes the enzyme.
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That's incorrect. Try again.

Incorrect. Review the following material: Cooperativity is a property of some enzymes with multiple subunits. It is triggered by the binding of a substrate molecule at one active site on the enzyme.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 42.1Compare and contrast the structure and functions of the major classes of circulatory systems in animals.
- 42.2Describe the role of the major structures of the heart during contraction and relaxation.
- 42.3Identify the types of blood vessels and their functions in maintaining and regulating blood flow.
- 42.4List the components of mammalian blood and their functions in exchange, transport, and defense.
- 42.5Compare and contrast the respiratory systems of aquatic animals, insects, and mammals.
- 42.6Describe the mechanisms for ventilating bird and human lungs, as well as the feedback pathways regulating human breathing.
- 42.7Use examples to describe the adaptive properties of respiratory pigments in gas exchange between body cells and the environment.